UML CLASS DIAGRAMS

Class → Attributes → Methods
Relationships → Multiplicity → Final Diagram

What will we learn today?

- What is a UML class diagram?
- What are class, object, attribute and method?
- How do we show relationships between classes?
- What does multiplicity mean?
- What are aggregation, composition and inheritance?
- How do we make a class diagram from YOUR project title?

1. What is a UML Class Diagram?

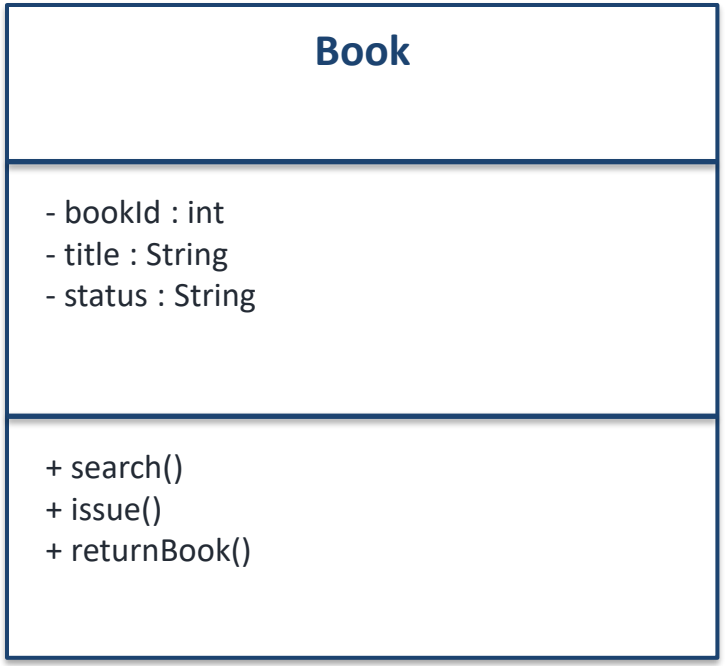
A class diagram shows the structure of a software system.

It tells us what important things/classes are in the system.

It shows what data each class stores.

It shows what work each class can do.

It also shows how classes are connected.



2. Class, Object and Attribute

Class = blueprint or design.

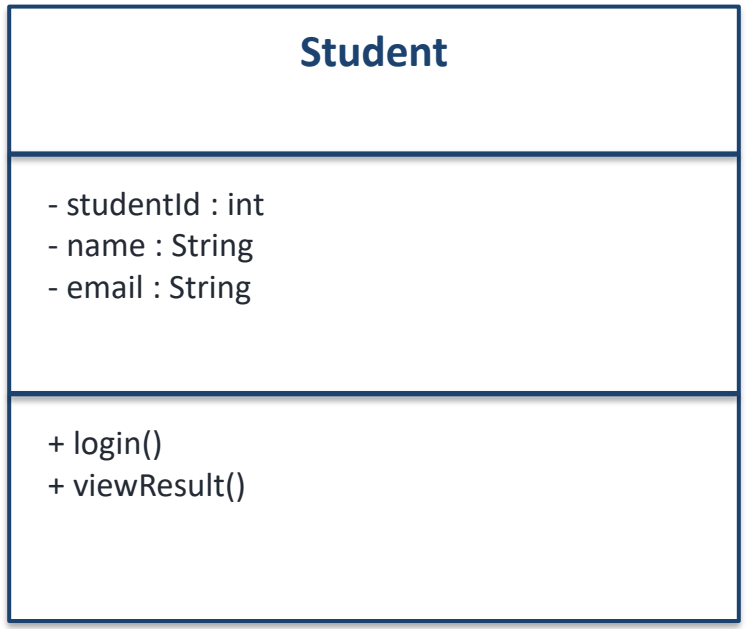
Object = one real instance of a class.

Attribute = data stored in the class.

Example: Book is a class.

bookId, title and author are attributes.

Book #101 can be one object.

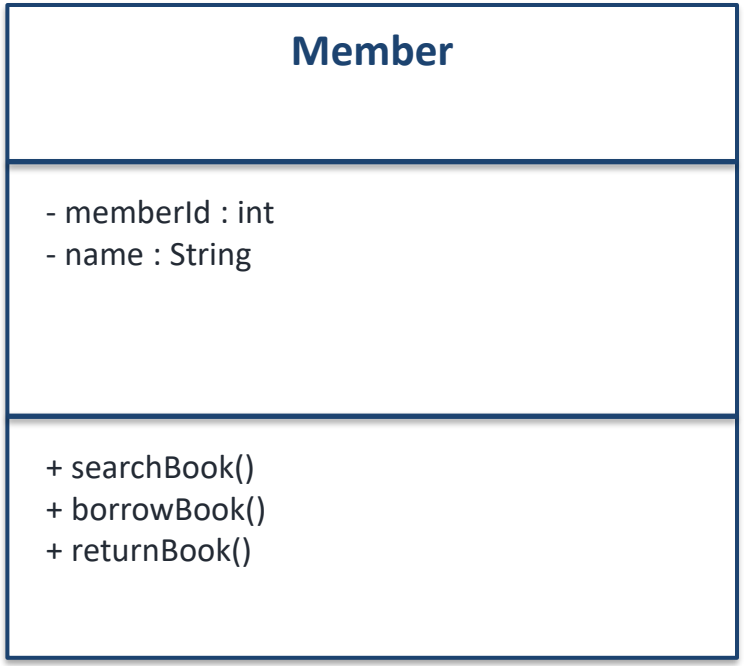


3. Operations / Methods

A method tells what a class can DO.

Usually written as an action/verb.

Examples: login(), addBook(), makePayment().



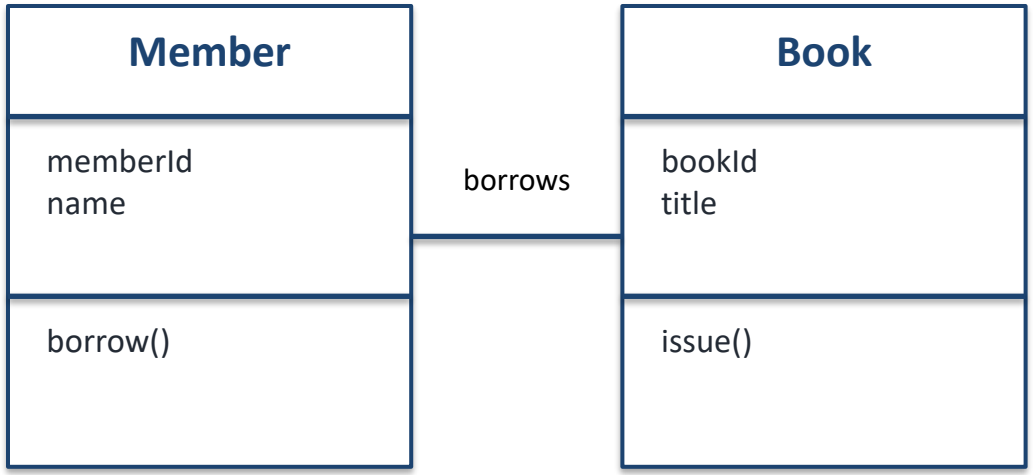
4. Association

Association means two classes are connected.

Example: Member borrows Book.

Draw a simple line between the classes.

You may write a small relationship name on the line.



5. Multiplicity: How Many?

1 = exactly one

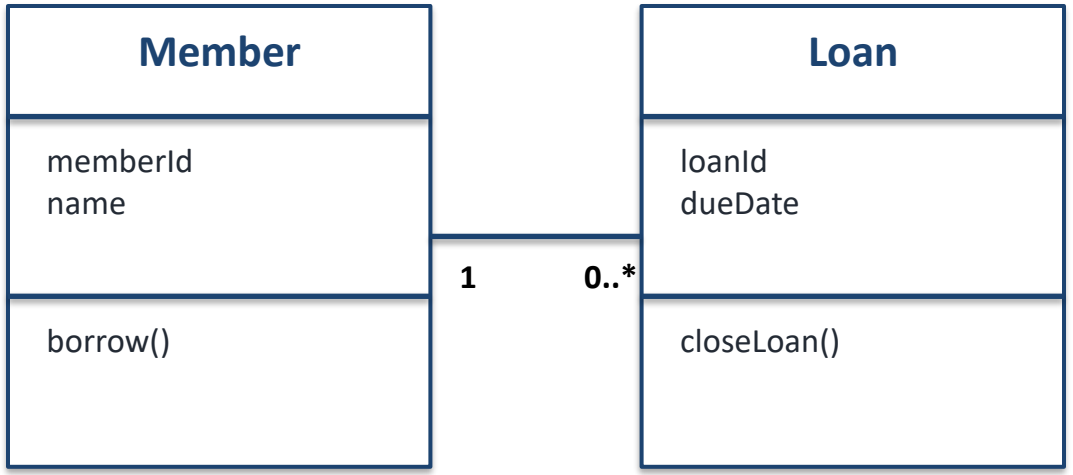
0..1 = zero or one

* = many

1..* = one or many

Example: one Member can have many Loan records.

Each Loan belongs to one Member.



6. Aggregation and Composition

Both show a whole-part relationship.

Aggregation = weak ownership.

The part can exist separately.

Composition = strong ownership.

If the whole is removed, the part normally does not exist independently.

Use them only when the whole-part meaning is clear.

Aggregation

Department ◇— Teacher

Composition

Order ◆— OrderItem

7. Inheritance

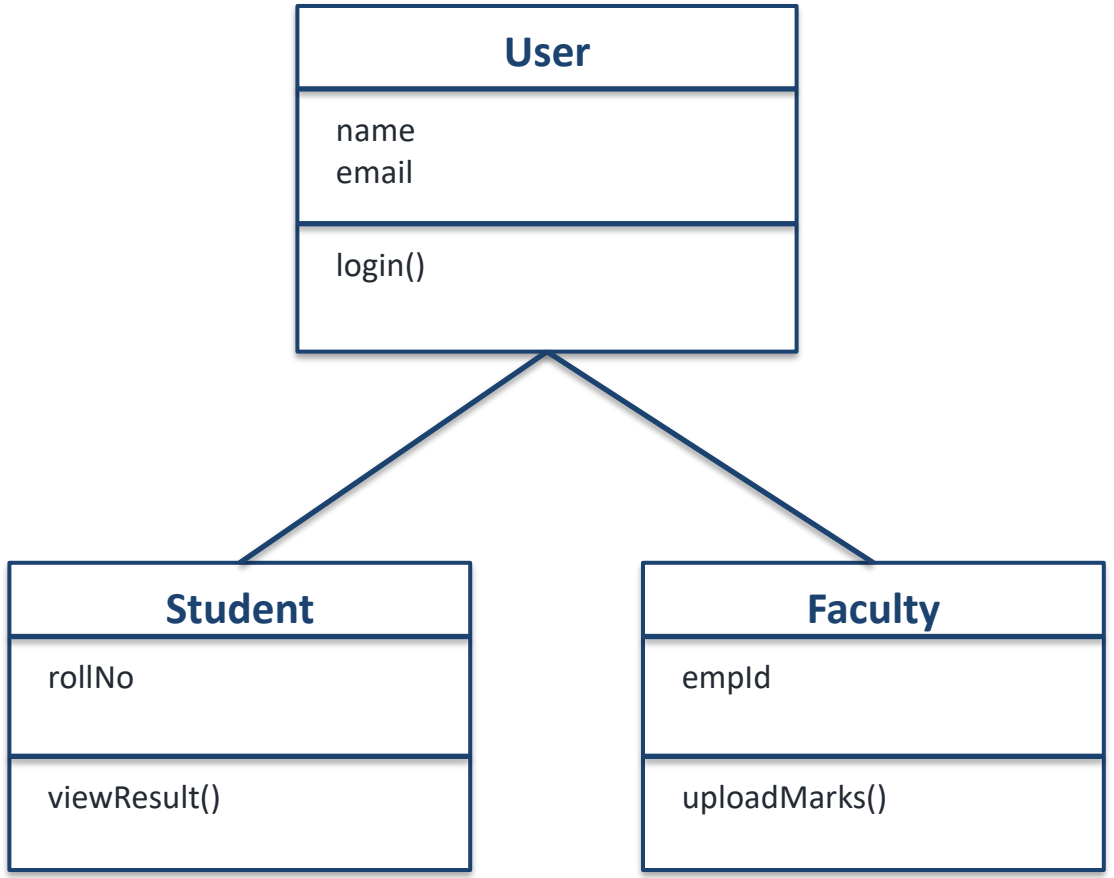
Inheritance means an IS-A relationship.

A child class gets common features from a parent class.

Example: Student is a User.

Faculty is also a User.

Do not use inheritance only because two classes look similar.



8. How to Find Classes from a Project Title

Step 1: Write your project title.

Step 2: Write 2–3 lines about what the system does.

Step 3: Circle important nouns: Student, Course, Order, Doctor, Vehicle, etc.

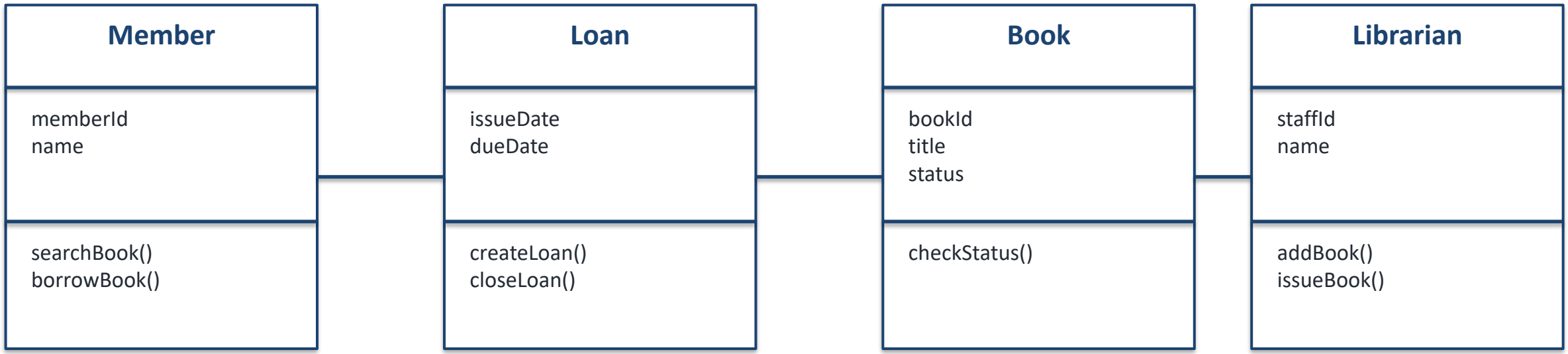
Step 4: Keep only nouns that need their own data or behaviour.

Step 5: Add 2–5 useful attributes to each class.

Step 6: Add 1–4 important methods.

Step 7: Connect classes and write multiplicity.

Worked Example: Library Management System



Think first about the real project: Member borrows Book, Loan stores issue details, Librarian manages books.

Now Apply It to YOUR Project

Project Title: _____

Main users/actors: _____

Possible classes: _____

For each class, write attributes and methods.

Decide relationships: association / aggregation / composition / inheritance.

Add multiplicity: 1, 0..1, *, or 1..*.

Finally, check whether the diagram matches your project requirements.

Common Mistakes

Suggested teaching time: 4 minutes

Do not convert every noun into a class.

Do not forget multiplicity.

Do not use inheritance unless there is a clear IS-A relationship.

Do not put too many attributes or methods.

Do not copy a ready-made diagram that does not match your project.

Keep names simple and readable.

5-Minute Classroom Activity

Suggested teaching time: 5 minutes

1. Write your group project title.
2. Identify any 4–6 possible classes.
3. Pick one class and write 3 attributes.
4. Write 2 methods for that class.
5. Show one relationship with multiplicity.

Summary + Link to Lab 5

Class = important entity in the system.

Attributes = data.

Methods = actions.

Relationships connect classes.

Multiplicity tells how many objects participate.